Section 6.1

Problem 2

The sample HDL levels (mg/dl) are

$$35,\ 49,\ 52,\ 54,\ 65,\ 51,\ 51,\ 47,\ 86,\ 36,\ 46,\ 33,\ 39,\ 45,\ 39,\ 63,\ 95,\ 35,\ 30,\ 48.$$

Thus,

$$n = 20, \qquad \sum x_i = 999.$$

(a) The point estimate of the population mean is the sample mean:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{999}{20} = 49.95.$$

Therefore, the point estimate of the population mean HDL level is

$$\bar{x} = 49.95 \text{ mg/dl}.$$

(b) The value separating the largest 50% from the smallest 50% is estimated by the sample median.

Ordering the data gives

$$30,\ 33,\ 35,\ 35,\ 36,\ 39,\ 39,\ 45,\ 46,\ 47,\ 48,\ 49,\ 51,\ 51,\ 52,\ 54,\ 63,\ 65,\ 86,\ 95.$$

Since $n = 20$ is even,

$$\tilde{x} = \frac{x_{(10)} + x_{(11)}}{2} = \frac{47 + 48}{2} = 47.5.$$

So the point estimate is

$$47.5 \text{ mg/dl}.$$

(c) The point estimate of the population standard deviation is the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Using $\bar{x} = 49.95$,

$$\sum_{i=1}^{20}(x_i - \bar{x})^2 = 5368.95.$$

Hence,

$$s = \sqrt{\frac{5368.95}{19}} = \sqrt{282.5763} \approx 16.81.$$

Thus, the point estimate of the population standard deviation is

$$16.81 \text{ mg/dl}.$$

(d) The point estimate of the population proportion p with HDL level at least 60 is the sample proportion:

$$\hat{p} = \frac{x}{n}.$$

The observations at least 60 are

$$63,\ 65,\ 86,\ 95,$$

so $x = 4$. Therefore,

$$\hat{p} = \frac{4}{20} = 0.20.$$

Problem 8

A sample of 80 components contains 12 defective components, so the number of nondefective components is

$$80 - 12 = 68.$$

(a) If p denotes the proportion of all such components that are not defective, then the point estimate of p is

$$\hat{p} = \frac{68}{80} = 0.85.$$

(b) For a series system to work properly, both components must work properly. Thus,

$$P(\text{system works}) = p^2.$$

Using the plug-in estimate,

$$\widehat{P(\text{system works})} = (\hat{p})^2 = (0.85)^2 = 0.7225.$$

Problem 10

Let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Each X_i has mean μ and variance σ^2, and the observations are independent.

(a) Using the identity $E(Y^2) = V(Y) + [E(Y)]^2$ with $Y = \bar{X}$,

$$E(\bar{X}^2) = V(\bar{X}) + [E(\bar{X})]^2.$$

Now

$$E(\bar{X}) = \mu, \qquad V(\bar{X}) = \frac{\sigma^2}{n},$$

so

$$E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2.$$

Since this is not equal to μ^2, $\bar{X}^2$ is not an unbiased estimator of μ^2.

(b) Consider the estimator $\bar{X}^2 - kS^2$. Then

$$E(\bar{X}^2 - kS^2) = E(\bar{X}^2) - kE(S^2).$$

From part (a),

$$E(\bar{X}^2) = \mu^2 + \frac{\sigma^2}{n},$$

and since S^2 is unbiased for σ^2,

$$E(S^2) = \sigma^2.$$

Therefore,

$$E(\bar{X}^2 - kS^2) = \mu^2 + \frac{\sigma^2}{n} - k\sigma^2 = \mu^2 + \sigma^2\left(\frac{1}{n} - k\right).$$

For unbiasedness, set this equal to μ^2:

$$\frac{1}{n} - k = 0.$$

Hence,

$$k = \frac{1}{n}.$$

So an unbiased estimator of μ^2 is

$$\bar{X}^2 - \frac{1}{n}S^2.$$

Section 6.2

Problem 22

Given

$$f(x;\theta) = \begin{cases} (\theta+1)x^\theta, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases} \qquad \theta > -1,$$

with sample

$x_1 = .92,\ x_2 = .79,\ x_3 = .90,\ x_4 = .65,\ x_5 = .86,\ x_6 = .47,\ x_7 = .73,\ x_8 = .97,\ x_9 = .94,\ x_{10} = .77.$

The sample mean is

$$\bar{x} = \frac{.92 + .79 + .90 + .65 + .86 + .47 + .73 + .97 + .94 + .77}{10} = \frac{8.00}{10} = 0.80.$$

(a) For the method of moments, compute $E(X)$:

$$E(X) = \int_0^1 x(\theta+1)x^\theta\,dx = (\theta+1)\int_0^1 x^{\theta+1}\,dx = (\theta+1)\cdot\frac{1}{\theta+2} = \frac{\theta+1}{\theta+2}.$$

Set $E(X) = \bar{x}$:

$$\frac{\theta+1}{\theta+2} = \bar{x}.$$

Solving for θ gives

$$\theta + 1 = \bar{x}(\theta+2),$$

$$\theta(1-\bar{x}) = 2\bar{x} - 1,$$

so

$$\hat{\theta}_{\text{MOM}} = \frac{2\bar{x}-1}{1-\bar{x}}.$$

Substituting $\bar{x} = 0.80$,

$$\hat{\theta}_{\text{MOM}} = \frac{2(0.80)-1}{1-0.80} = \frac{0.60}{0.20} = 3.$$

(b) The likelihood function is

$$L(\theta) = \prod_{i=1}^{10}(\theta+1)x_i^\theta = (\theta+1)^{10}\prod_{i=1}^{10}x_i^\theta.$$

Thus the log-likelihood is

$$\ell(\theta) = 10\ln(\theta+1) + \theta\sum_{i=1}^{10}\ln x_i.$$

Differentiate and set equal to 0:

$$\ell'(\theta) = \frac{10}{\theta+1} + \sum_{i=1}^{10}\ln x_i = 0.$$

Hence,

$$\hat{\theta}_{\text{MLE}} = -\frac{10}{\sum_{i=1}^{10}\ln x_i} - 1.$$

Now

$$\sum_{i=1}^{10}\ln x_i = \ln(.92)+\ln(.79)+\ln(.90)+\ln(.65)+\ln(.86)+\ln(.47)+\ln(.73)+\ln(.97)+\ln(.94)+\ln(.77) \approx -2.4295.$$

Therefore,

$$\hat{\theta}_{\text{MLE}} = -\frac{10}{-2.4295} - 1 \approx 3.1161.$$

Also,

$$\ell''(\theta) = -\frac{10}{(\theta+1)^2} < 0,$$

so this critical point gives a maximum.

Problem 24

Let X be the number of incorrect diagnoses before the rth correct diagnosis. Here,

$$r = 3, \qquad x = 17,$$

since there were 20 diagnoses in all and $20 - 3 = 17$ of them were incorrect.
For a negative binomial random variable,

$$P(X = x) = \binom{x+r-1}{x} p^r (1-p)^x, \qquad x = 0, 1, 2, \ldots$$

so the likelihood function is

$$L(p) = \binom{19}{17} p^3 (1-p)^{17}, \qquad 0 < p < 1.$$

Taking logs,

$$\ell(p) = \ln\binom{19}{17} + 3\ln p + 17\ln(1-p).$$

Differentiate and set equal to 0:

$$\ell'(p) = \frac{3}{p} - \frac{17}{1-p} = 0.$$

Thus,

$$3(1-p) = 17p,$$

so

$$3 = 20p, \qquad \hat{p}_{\text{MLE}} = \frac{3}{20} = 0.15.$$

If a random sample of 20 mechanics results in 3 correct diagnoses, the binomial likelihood is proportional to

$$p^3(1-p)^{17},$$

which is the same function of p. Therefore, the mle is again

$$\hat{p}_{\text{MLE}} = \frac{3}{20} = 0.15.$$

So the two mles are the same.
From Exercise 17, the unbiased estimator is

$$\hat{p} = \frac{r-1}{X+r-1}.$$

With $r = 3$ and $X = 17$,

$$\hat{p}_{\text{unb}} = \frac{3-1}{17+3-1} = \frac{2}{19} \approx 0.1053.$$

Hence,

$$\hat{p}_{\text{MLE}} = 0.15 > 0.1053 = \hat{p}_{\text{unb}}.$$

Section 7.1

Problem 2

The two confidence intervals for μ are

$$(114.4, 115.6) \qquad \text{and} \qquad (114.1, 115.9).$$

(a) The sample mean is the midpoint of either interval:

$$\bar{x} = \frac{114.4 + 115.6}{2} = 115.0$$

and also

$$\bar{x} = \frac{114.1 + 115.9}{2} = 115.0.$$

Thus,

$$\bar{x} = 115.0 \text{ Hz}.$$

(b) Both intervals are based on the same sample data, so they have the same center. The interval with the higher confidence level must be wider.

The widths are

$$115.6 - 114.4 = 1.2$$

and

$$115.9 - 114.1 = 1.8.$$

Thus $(114.1, 115.9)$ is the wider interval, so it is the 99% interval. Therefore,

$$(114.4, 115.6)$$

is the 90% confidence interval.

Problem 4

Given

$$\sigma = 3.0, \qquad \bar{x} = 58.3.$$

Since σ is known, the confidence interval for μ is

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

(a) For a 95% confidence interval with $n = 25$,

$$E = 1.96\left(\frac{3}{\sqrt{25}}\right) = 1.176.$$

Therefore,

$$58.3 \pm 1.176 = (57.124,\ 59.476) \approx (57.12,\ 59.48).$$

(b) For a 95% confidence interval with $n = 100$,

$$E = 1.96\left(\frac{3}{\sqrt{100}}\right) = 0.588.$$

Therefore,

$$58.3 \pm 0.588 = (57.712,\ 58.888) \approx (57.71,\ 58.89).$$

(c) For a 99% confidence interval with $n = 100$,

$$E = 2.576\left(\frac{3}{10}\right) = 0.7728.$$

Therefore,

$$58.3 \pm 0.7728 = (57.5272,\ 59.0728) \approx (57.53,\ 59.07).$$

(d) For an 82% confidence interval, $\alpha = 0.18$, so $\alpha/2 = 0.09$ and

$$z_{0.09} = z_{0.91} \approx 1.341.$$

With $n = 100$,

$$E = 1.341\left(\frac{3}{10}\right) = 0.4023.$$

Therefore,

$$58.3 \pm 0.4023 = (57.8977,\ 58.7023) \approx (57.90,\ 58.70).$$

(e) The width of a 99% confidence interval is

$$2z_{0.005}\frac{\sigma}{\sqrt{n}}.$$

Set this equal to 1.0:

$$2(2.576)\frac{3}{\sqrt{n}} = 1.$$

Then

$$\sqrt{n} = 2(2.576)(3) = 15.456,$$

so

$$n = (15.456)^2 \approx 238.89.$$

Rounding up,

$$n = 239.$$

Problem 10

The sample lifetimes (years) are

$$2.0,\ 1.3,\ 6.0,\ 1.9,\ 5.1,\ 0.4,\ 1.0,\ 5.3,\ 15.7,\ 0.7,\ 4.8,\ 0.9,\ 12.2,\ 5.3,\ 0.6.$$

Thus,

$$n = 15, \qquad \sum x_i = 63.2.$$

Assume the lifetime distribution is exponential with rate parameter λ. Then

$$\mu = \frac{1}{\lambda}$$

and

$$2\lambda \sum_{i=1}^{n} X_i \sim \chi^2_{2n}.$$

Since $2n = 30$,

$$2\lambda \sum X_i \sim \chi^2_{30}.$$

(a) For a 95% confidence interval,

$$P\left(\chi^2_{.975,30} < 2\lambda \sum X_i < \chi^2_{.025,30}\right) = 0.95.$$

Using

$$\chi^2_{.025,30} = 46.979, \qquad \chi^2_{.975,30} = 16.791,$$

we obtain

$$P\left(\frac{16.791}{2\sum X_i} < \lambda < \frac{46.979}{2\sum X_i}\right) = 0.95.$$

Since $\mu = 1/\lambda$,

$$P\left(\frac{2\sum X_i}{46.979} < \mu < \frac{2\sum X_i}{16.791}\right) = 0.95.$$

Substituting $\sum X_i = 63.2$,

$$\left(\frac{126.4}{46.979}, \frac{126.4}{16.791}\right) = (2.691,\ 7.528) \approx (2.69,\ 7.53).$$

(b) For a 99% confidence interval,

$$P\left(\chi^2_{.995,30} < 2\lambda \sum X_i < \chi^2_{.005,30}\right) = 0.99.$$

Using

$$\chi^2_{.005,30} = 53.672, \qquad \chi^2_{.995,30} = 13.787,$$

we get

$$P\left(\frac{2\sum X_i}{53.672} < \mu < \frac{2\sum X_i}{13.787}\right) = 0.99.$$

Thus,

$$\left(\frac{126.4}{53.672}, \frac{126.4}{13.787}\right) = (2.355,\ 9.168) \approx (2.36,\ 9.17).$$

(c) For an exponential distribution,

$$\sigma = \frac{1}{\lambda} = \mu.$$

Therefore, the 95% confidence interval for the population standard deviation is the same as the 95% confidence interval for the population mean from part (a):

$$(2.69,\ 7.53) \text{ years}.$$

Section 7.2

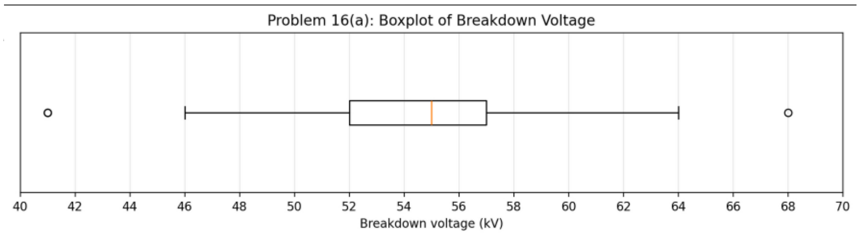
Problem 16

Using all 72 observations from the two tables, the ordered-data summary is

$$\min = 41, \qquad Q_1 = 52, \qquad \tilde{x} = 55, \qquad Q_3 = 57, \qquad \max = 68.$$

Thus,

$$\text{IQR} = Q_3 - Q_1 = 57 - 52 = 5.$$

(a) 

The boxplot has lower whisker at 46, upper whisker at 64, and box from 52 to 57 with median at 55.

The distribution is centered near 55 kV, with moderate spread and three outliers: two low outliers at 41 and one high outlier at 68.

(b) For the 72 observations,

$$\bar{x} = 54.444, \qquad s = 5.040.$$

Since $n = 72$ is large, a 95% confidence interval for μ is

$$\bar{x} \pm z_{0.025}\frac{s}{\sqrt{n}} = 54.444 \pm 1.96\frac{5.040}{\sqrt{72}}.$$

The margin of error is

$$1.96\frac{5.040}{\sqrt{72}} \approx 1.164.$$

Therefore,

$$54.444 \pm 1.164 = (53.280,\ 55.609) \approx (53.28,\ 55.61).$$

We are 95% confident that the true average breakdown voltage lies between 53.28 and 55.61 kV. The interval width is about 2.33 kV, so μ appears to have been estimated fairly precisely.

(c) If virtually all values lie between 40 and 70, then using the empirical rule,

$$\mu \pm 2\sigma \approx 40 \text{ to } 70.$$

Hence,

$$4\sigma \approx 70 - 40 = 30, \qquad \sigma \approx 7.5.$$

For a 95% confidence interval with width 2, the margin of error is 1, so

$$1 = 1.96\frac{7.5}{\sqrt{n}}.$$

Thus,

$$\sqrt{n} = 14.7, \qquad n = (14.7)^2 = 216.09.$$

Rounding up,

$$n = 217.$$

Problem 20

Given

$$\hat{p} = 0.53, \qquad n = 2343.$$

For a confidence interval for a population proportion,

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

(a) For a 99% confidence interval, $z_{0.005} = 2.576$. Thus,

$$E = 2.576\sqrt{\frac{0.53(0.47)}{2343}} \approx 0.0266.$$

Therefore, the 99% confidence interval is

$$0.53 \pm 0.0266 = (0.5034,\ 0.5566) \approx (0.503,\ 0.557).$$

We are 99% confident that the true proportion of all adult Americans who had watched digitally streamed TV programming up to that time lies between 0.503 and 0.557.

(b) To make the width at most 0.05, the margin of error must satisfy

$$E \leq 0.025.$$

To guarantee this regardless of the value of $\hat{p}$, use the largest possible value of $\hat{p}(1-\hat{p})$, namely 0.25. Then

$$0.025 = 2.576\sqrt{\frac{0.25}{n}}.$$

Solving for n,

$$n = \frac{0.25(2.576)^2}{(0.025)^2} \approx 2653.59.$$

Rounding up,

$$n = 2654.$$

Problem 26

From the frequency table,

$$n = 50$$

and

$$\sum x_i f_i = 0(1) + 1(4) + 2(8) + 3(10) + 4(8) + 5(7) + 6(5) + 7(3) + 8(2) + 9(1) + 10(1) = 203.$$

So

$$\bar{x} = \frac{203}{50} = 4.06.$$

For a Poisson distribution,

$$E(X) = \mu, \qquad V(X) = \mu,$$

so for large n,

$$Z = \frac{\bar{X} - \mu}{\sqrt{\mu/n}}$$

is approximately standard normal. For a 95% confidence interval,

$$P\left(-1.96 \le \frac{\bar{X} - \mu}{\sqrt{\mu/n}} \le 1.96\right) \approx 0.95.$$

Using $\bar{x} = 4.06$, solve

$$|\bar{x} - \mu| \le 1.96\sqrt{\frac{\mu}{50}}.$$

Squaring both sides gives

$$(\bar{x} - \mu)^2 \le \frac{(1.96)^2\mu}{50},$$

so

$$50(\mu - 4.06)^2 \le 3.8416\mu.$$

Expanding,

$$50\mu^2 - 409.8416\mu + 824.18 \le 0.$$

The endpoints are the roots of

$$50\mu^2 - 409.8416\mu + 824.18 = 0,$$

which are

$$\mu \approx 3.5386 \qquad \text{and} \qquad \mu \approx 4.6582.$$

Therefore, the large-sample 95% confidence interval for μ is

$$(3.54,\ 4.66).$$

Section 7.3

Problem 30

For a two-sided confidence interval with confidence level $1 - \alpha$, the critical value is

$$t_{\alpha/2,\,\nu}.$$

(a) Confidence level 95%, $\nu = 10$:

$$t_{0.025,10} \approx 2.228.$$

(b) Confidence level 95%, $\nu = 15$:

$$t_{0.025,15} \approx 2.131.$$

(c) Confidence level 99%, $\nu = 15$:

$$t_{0.005,15} \approx 2.947.$$

(d) Confidence level 99%, $n = 5$, so $\nu = n - 1 = 4$:

$$t_{0.005,4} \approx 4.604.$$

(e) Confidence level 98%, $\nu = 24$:

$$t_{0.01,24} \approx 2.492.$$

(f) Confidence level 99%, $n = 38$, so $\nu = n - 1 = 37$:

$$t_{0.005,37} \approx 2.715.$$

Problem 34

Given

$$n = 14, \qquad \bar{x} = 8.48 \text{ MPa}, \qquad s = 0.79 \text{ MPa}.$$

Then $df = n - 1 = 13$. For a one-sided 95% bound,

$$t_{0.05,13} \approx 1.771.$$

(a) A 95% lower confidence bound for the true mean μ is

$$\bar{x} - t_{0.05,13}\frac{s}{\sqrt{n}}.$$

Substituting,

$$8.48 - 1.771\left(\frac{0.79}{\sqrt{14}}\right).$$

Since

$$\frac{0.79}{\sqrt{14}} \approx 0.2111,$$

the lower bound is

$$8.48 - 1.771(0.2111) \approx 8.1061 \approx 8.11.$$

Thus, the 95% lower confidence bound is 8.11 MPa. This means there is 95% confidence that the true average proportional limit stress is at least 8.11 MPa.

This procedure assumes the sample is random and independent and that the population distribution is approximately normal.

(b) A 95% lower prediction bound for the proportional limit stress of a single future joint is

$$\bar{x} - t_{0.05,13} s\sqrt{1 + \frac{1}{n}}.$$

Thus,

$$8.48 - 1.771(0.79)\sqrt{1 + \frac{1}{14}}.$$

Since

$$\sqrt{1 + \frac{1}{14}} = \sqrt{\frac{15}{14}} \approx 1.0351,$$

the lower prediction bound is

$$8.48 - 1.771(0.79)(1.0351) \approx 7.0319 \approx 7.03.$$

So the 95% lower prediction bound is 7.03 MPa.

Problem 36

Given

$$n = 26, \qquad \bar{x} = 370.69, \qquad s = 24.36.$$

Since the normal probability plot shows a substantial linear pattern, a one-sample t procedure is appropriate.
For a one-sided upper confidence bound with confidence level 95%,

$$\bar{x} + t_{0.05,25} \frac{s}{\sqrt{n}}.$$

From the t table,

$$t_{0.05,25} \approx 1.708.$$

Therefore,

$$370.69 + 1.708\left(\frac{24.36}{\sqrt{26}}\right).$$

Since

$$\frac{24.36}{\sqrt{26}} \approx 4.7774,$$

the upper confidence bound is

$$370.69 + 1.708(4.7774) \approx 378.85.$$

Thus, the 95% upper confidence bound for the population mean escape time is

$$378.85 \text{ sec}.$$

Section 7.4

Problem 42

Using chi-square critical values with right-tail area α:

(a)
$$\chi^2_{.1,15} \approx 22.307.$$

(b)
$$\chi^2_{.1,25} \approx 34.382.$$

(c)
$$\chi^2_{.01,25} \approx 44.314.$$

(d)
$$\chi^2_{.005,25} \approx 46.928.$$

(e)
$$\chi^2_{.99,25} \approx 11.524.$$

(f)
$$\chi^2_{.995,25} \approx 10.520.$$

Problem 44

Given
$$n = 9, \qquad s = 2.81, \qquad s^2 = (2.81)^2 = 7.8961.$$
Assuming normality, a 95% confidence interval for σ^2 is
$$\left(\frac{(n-1)s^2}{\chi^2_{.025,\,n-1}},\ \frac{(n-1)s^2}{\chi^2_{.975,\,n-1}} \right).$$
Here,
$$n - 1 = 8, \qquad (n-1)s^2 = 8(7.8961) = 63.1688.$$
From the chi-square table,
$$\chi^2_{.025,8} = 17.535, \qquad \chi^2_{.975,8} = 2.180.$$
Therefore,
$$\left(\frac{63.1688}{17.535},\ \frac{63.1688}{2.180} \right) = (3.603,\ 28.977) \approx (3.60,\ 28.98).$$
So the 95% confidence interval for σ^2 is
$$(3.60,\ 28.98).$$
Taking square roots of the endpoints,
$$\left(\sqrt{3.603},\ \sqrt{28.977} \right) = (1.898,\ 5.383) \approx (1.90,\ 5.38).$$
Thus, the 95% confidence interval for σ is
$$(1.90,\ 5.38).$$

Problem 46

The sample observations ($\mathrm{kN/m^2}$) are

$$33.2,\ 41.8,\ 37.3,\ 40.2,\ 36.7,\ 39.1,\ 36.2,\ 41.8,\ 36.0,\ 35.2,\ 36.7,\ 38.9,\ 35.8,\ 35.2,\ 40.1.$$

Thus,

$$n = 15.$$

The sample mean and sample standard deviation are

$$\bar{x} = \frac{564.2}{15} = 37.613, \qquad s \approx 2.572,$$

so

$$s^2 \approx 6.613.$$

(a) The ordered data are

$$33.2,\ 35.2,\ 35.2,\ 35.8,\ 36.0,\ 36.2,\ 36.7,\ 36.7,\ 37.3,\ 38.9,\ 39.1,\ 40.1,\ 40.2,\ 41.8,\ 41.8.$$

They are reasonably symmetric, with no obvious extreme outliers or strong skewness. Thus, it is plausible that the sample came from a normal population.

(b) Assuming normality,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

For a 95% upper confidence bound for σ^2,

$$P\left(\frac{(n-1)S^2}{\sigma^2} \geq \chi^2_{.95,14}\right) = 0.95.$$

Solving for σ^2 gives

$$P\left(\sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{.95,14}}\right) = 0.95.$$

Therefore, the 95% upper confidence bound for σ is

$$U_\sigma = \sqrt{\frac{(n-1)s^2}{\chi^2_{.95,14}}}.$$

Using

$$n-1 = 14, \qquad s^2 \approx 6.613, \qquad \chi^2_{.95,14} \approx 6.571,$$

we get

$$U_\sigma = \sqrt{\frac{14(6.613)}{6.571}} = \sqrt{14.090} \approx 3.754.$$

So the 95% upper confidence bound for the population standard deviation is

$$3.75\ \mathrm{kN/m^2}.$$